

Using machine learning to create a credit scoring model in banking and finance

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Abstract — Recently, personal credit is one of the most important products among all credit products that banks offer on the market. Banks and Financial Institutions (FIs) are always trying to find an effective credit score assessment model to reduce lending risks as well as increase income for the banks and financial institutions. In this paper, machine learning is applied to create credit scoring model for bank application. The work also discussed on how to calculate and set threshold for setting an ideal credit score cut-off point. Experimental results show that our proposed method can be applied in the banks or Credit Institutions to reduce risks in loan services.

Keywords— Credit scoring, scorecard model, weight of evidence, logistic regression, lightGBM, support vector machine.

I. INTRODUCTION

In the process of globalization, international economic integration requires the bank system to play a big role in the economic and financial relationship. Among many operations that banks conduct, lending and deposit are the biggest two with the highest ratio in the source of capital and total assets of banks [1]. Despite the fact that it has always been a critical business operation, lending, simultaneously, is a damaging threat, affects to banks' operations seriously. And among all credit products that banks offer on the market, personal credit is irreplaceable, since personal loans account for a big ratio in the total loan amount.

In 2020, credit growth in Vietnam reached 12.1 per cent, which was the lowest growth rate in the previous five years. In 2021, credit growth is expected to bounce back to around 14 per cent [2]. Given the fact that credit growth in Vietnam is considered rapid in the Southeast Asia region, credit risk cannot be effectively controlled by credit officers, at least in the traditional way they have been doing. Therefore, many credits scoring systems have been created to shorten this process. The proposed approach in this work aims to create a credit scoring model in banking and finance in order to support staff in the specific context of credit scoring.

In this study, a method for credit scoring based on various criteria are proposed and integrated with a logistic regression classifier in credit scoring tasks. In which, a threshold is calculated and set for setting an ideal credit score cut-off point.

The paper is organized as follows: In section 2, related work is introduced. Section 3 presents credit scorecard models. In section 4, the experiments and results are shown. Finally, in section 5, the conclusions are given.

II. RELATED WORK

X. Zhang et al. [3] explored different credit scoring models through machine learning in 2018. Most of them use neural networks and support vector machine (SVM) models. Due to the presence of relatively large amounts of data and some

redundant features present in credit data, it is not always possible to make accurate predictions. In this paper, the authors proposed a credit model named novel credit scoring model (NCSM) that reduces the amount of redundant and unrelated data using Optimized Random Forest. The authors proposed a smaller number of features that would reduce the work of the credit assessors. The authors intend to continue research on credit structure data structures and the use of the grouping approach by the random forest method.

In another method presented in 2018, C. Wang et al. [4] tried to find a correct and relevant credit model for peer – to – peer loans. The authors proposed the model of interlocutory lending, which is growing faster in China due to the simple lending process. Although the lending process is easy, the associated risks for lenders are high. The authors used the online behavioral data of borrowers and offered a credit score that uses machine learning. They used the short-term memory technique (LSTM). The authors claimed that research done with databases using LSTM showed better results. In future research, the authors intend to add the time interval of each event.

In a research study published in 2017, G. Arutjothi and C. Senthamarai [5] aimed to predict the status of loans at commercial banks using machine learning techniques. The authors wanted to build a credit rating model that banks can use to extend credit. From previous research articles, the authors found that there is no model sacred. The authors used an analytical model for credit data. This is done by a combination of min-max normalization and Nearest Neighbor (K-NN) classifier. The authors showed that the proposed model can help build a tool that can predict the status of loans in commercial banks. Besides, the proposed model achieved an accuracy of 75% and can be improved by further studies.

III. CREDIT SCORECARD MODELS

A. Credit decision process

Credit scoring is usually applied to relatively small loans offered by commercial banks and financial institutions. Borrowers will not have to use an asset as collateral, so they probably need a high credit score to get a good interest, the term “unsecured” is to distinguish this type of loan from mortgage loan, which is defined as “secured”. Normally, a loan application goes through a set of steps, which is proceeded by many conductors and departments as shown in Fig 1. Thus, it takes quite a while for banks to disburse each personal loan due to no collateral.

B. Credit scorecard model [6]

Financial institutions and commercial banks have complex credit models that use the information contained in data warehouse like salary, credit commitments and historical loan

data to determine a credit score of an application or an existing customer. The model generates a score that represents the probability that the lender will receive a repayment on time if they give a person a loan or credit card.

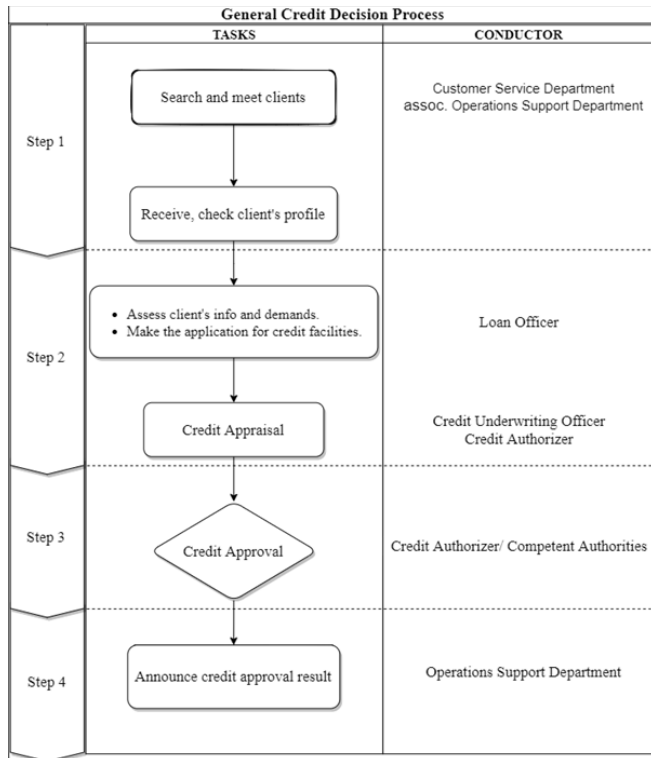


Fig. 1. General Credit Decision Process

A credit scorecard is one of those credit models, it is one of the most common credit models because it is relatively easy to interpret for customers and it has existed for the last few decades, so the development process it is standard and widely understood. Credit scorecards represent different characteristics of a customer (age, residency status, time at current address, time at current job, etc.) translated into points and the total number of points is converted into the credit score. Therefore, a credit scorecard is a search table that maps a borrower's specific characteristics in points. The total number of points is converted to a credit score.

For example, a credit card can give points to individual borrowers for their age and income according to Table I.

TABLE I. EXAMPLE OF A SCORECARD

Features	Values	Points
Age	Up to 25	10
	26 to 40	25
	41 to 65	38
	66 and up	43
Income	Up to 20 mil	16
	20 mil to 50 mil	28

Using the credit card in this example, a particular customer who is 31 years old and has an income of 30 million a year is in the second age class (26-40) and gets 25 points for his age,

and similarly, he gets 28 points for his income, which means the total for these two features is 53. In fact, there are still many other features to put into the scoring process, therefore, to shorten this calculation, building a scorecard model using machine learning algorithms is a bright idea.

C. Weight of Evidence (WoE)

WoE [7] is one of the most common feature engineering and feature selection methods which are applied in the scorecard model. Based on the ratio of good candidates to bad candidates at each group level, this method measures the "strength" of the pool to differentiate between good and bad risk and attempts to find a monotonous relationship between the independent variables and the target variable. The criterion for ranking is Information Value (IV), IV assists with ranking features based on their relative importance.

The formula to calculate WoE is as in the equation (1):

$$WoE = \ln \left(\frac{\% \text{ of good customers}}{\% \text{ of bad customers}} \right) \quad (1)$$

A positive WoE means that the proportion of good customers is more than that of bad customers and vice versa for a negative WoE value.

Information value (IV) is calculated as follows:

$$IV = \sum (\% \text{ of good customers} - \% \text{ of bad customers}) \times WoE \quad (2)$$

IV. EXPERIMENTS

A. Dataset

The work uses a database available on Kaggle that relates to consumer loans provided by Lending Club [8]. The raw data includes information on more than 450,000 consumer loans granted between 2007 and 2014 with almost 75 characteristics, including the current loan status and various attributes related to both borrowers and their repayment behavior. Table II below shows basic information of the dataset.

TABLE II. DATASET INFO

	Total Dataset
Number of observations	466,285
Number of features	74
Features with missing values	18

Initial data research reveals the followings:

- There are 18 features with more than 80% missing values, such as: mths_since_last_record, annual_inc_joint, dti_joint, verification_status_joint.... Given the high proportion of missing values, any technique to impute them is likely to result in inaccurate results.
- Certain static features not related to credit risk, e.g., id, member_id, url, title (the loan title provided by the borrower).
- (The job title supplied by the borrower when applying for the loan): the input values by borrowers are inconsistent and not yet verified, which would lead to inaccurate prediction.

- Other predictive functions that are expected to complete only after the borrower has failed, for example, bailouts, collection_recovery_. Because our goal here is to predict the future probability of a default, having such functions in our model will be counterintuitive as they will not be observed until the predetermined event occurs.

All the above features will be dropped due to for following reason as pointed out in Table III:

TABLE III. REASONS TO DROP FEATURES AFTER PRELIMINARY DATA EXPLORATION.

Feature Name	Reason to drop
18 missing features	Values missing as explained above.
<i>id</i> , <i>member_id</i> , <i>title</i> , <i>emp_titl'</i> , <i>url</i> , <i>desc</i> , <i>zip_code</i>	Redundant as explained above.
<i>recoveries</i> , <i>collection_recovery_fee</i> , <i>total_rec_prnc'</i> , <i>total_rec_late_fee</i>	Forward-looking as explained above.
<i>sub_grade</i>	Same information is captured in <i>grade</i> column.
<i>next_pymnt_d</i>	Data is historical and this column is supposed to have future dates, therefore, it will not make sense for the model.

After data cleaning and feature selection, the collected dataset contains 466,285 observations and 51 features with the statistic shown in Table IV. To validate the models, the 5-fold cross validation technique is applied.

TABLE IV. DATASET INFO AFTER PREPROCESSING

	Total Dataset
Number of observations	466,285
Number of features	51

B. Experiment results

The models we carried out in this research are: LightGBM, Logistic Regression (LG), and Support Vector Machine (SVM).

After performing 5-folds validation on the training set for the three models, we received results as shown in Table V. The results show that Logistic Regression returns a more impressive result.

TABLE V. EXPERIMENTAL ACCURACY

Model	LightGBM	LG	SVM
Fold 1	0.7727	0.865780	0.84116
Fold 2	0.7748	0.865808	0.84057
Fold 3	0.7736	0.865809	0.84051
Fold 4	0.7723	0.865821	0.84057
Fold 5	0.7737	0.865823	0.84053
Average	0.7734	0.865808	0.84067

C. Scorecard Development

Then we will determine the minimum and maximum points that our scorecard should yield. As a starting point, we will use the same range of scores that FICO uses: 300 to 850.

The coefficients returned by the logistic regression model for each feature category are then scaled to our range of credit scores using simple arithmetic. An additional step here is to update the model intercept credit score with an additional scale, which will then be used as the starting point for each score calculation.

The score calculation of each feature is the equation (3):

$$score = coefficient \times \frac{max\ score - min\ score}{\sum max\ coefficients - \sum min\ coefficients} \quad (3)$$

In this case, min score is 300 while max score is 850, based on the score range of FICO Scores [9] as shown in Fig.2.

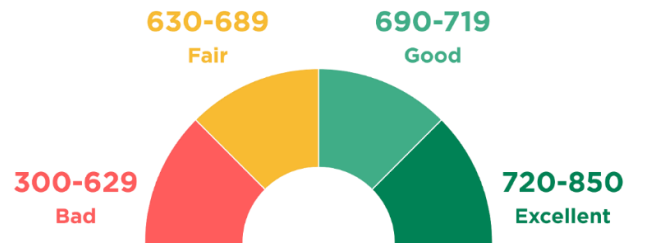


Fig. 2. The score range of FICO Scores [9].

After the scoring process, we investigated the distribution of credit scores on this data set as shown in Fig 3. The overall shape of the distribution is relatively skewed to the left with a clear mode around 550. In addition, it has another (smaller) “peak” (mode) around 500. The spread of the distribution is of course between the already set max and min score of 300 – 850. However, the actual max score of this dataset does not surpass 750.

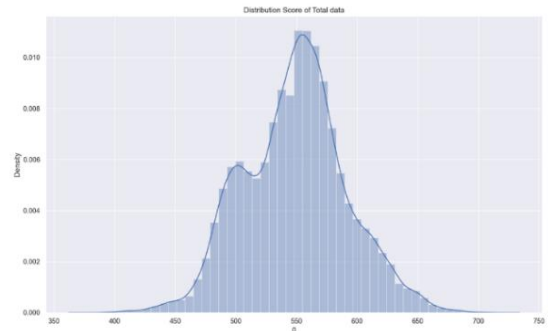


Fig. 3. Score distribution on total data.

D. Setting Loan Approval Cut-offs

While working with the dataset, based on the loans, the borrowers are classified into good customers and bad customers:

- Good customers: Current, Fully Paid, In Grace Period, Does not meet the credit policy. Status: Fully Paid, Late (16–30 days)
- Bad customers: Charged Off, Default, Late (31–120 days), Does not meet the credit policy. Status: Charged Off

As shown in Fig 4, the proportion of good customers is much higher than that of bad ones, which means imbalanced data. So, the need for a threshold where banks and FIs can make use to maximize the approval rate which is similar to conversion rate in sales is found. The higher the rate, the higher the cashflow.

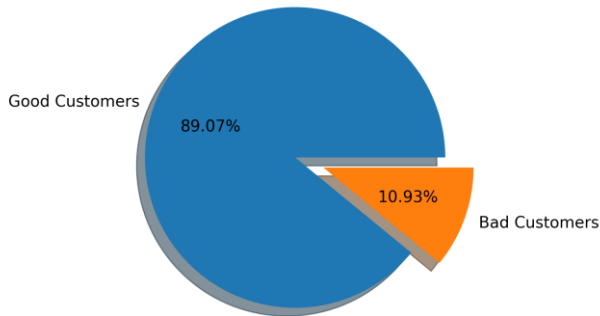


Fig. 4. Proportion of good and bad customers.

Given the demand above, it needs to find an ideal credit score cut-off point which means potential borrowers with a credit score higher than this cut-off point will be accepted and those less than it will be rejected to determine which loans to approve and reject. This ideal threshold is calculated using the Youden's J statistic [10]:

$$J = TPR - FPR \quad (4)$$

The Youden's index is often used in conjunction with receiver performance characteristic (ROC) analysis. The index is defined for all points on the ROC curve and the maximum value of the index can be used as a criterion for selecting the optimal cut – off point. ROC curves in logistic regression are used for determining the best cut – off value for predicting whether a new observation is a "failure" (0) or a "success" (1), or in this work is "good customer" or "bad customer". The ROC curve of the proposed model is shown in Fig 5:

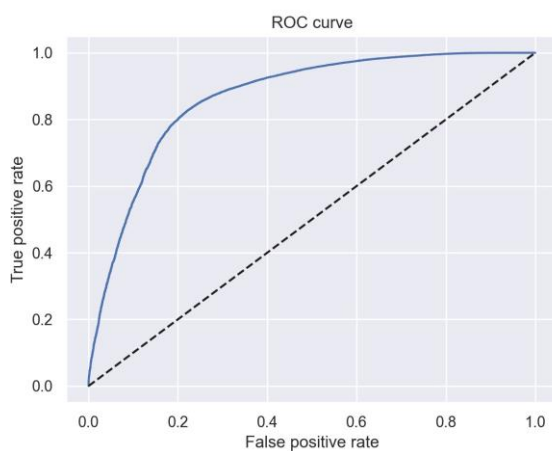


Fig. 5. ROC curve plot.

Based on the Youden's J statistic, the ideal probability threshold is 0.186472 which minimizes the False Positive

Rate (FPR) and maximizes the True Positive Rate (TPR) – all samples with a predicted probability higher than this should be classified as in Default and vice versa. This can be confirmed by comparing confusion matrices before and after applying the threshold using Logistic Regression in Table VI and Table VII:

TABLE VI. CONFUSION MATRIX BEFORE APPLYING THRESHOLD

	Predicted as Positive	Predicted as Negative
Positive	0.78993053	0.21006947
Negative	0.1910928	0.8089072

TABLE VII. CONFUSION MATRIX AFTER APPLYING THRESHOLD

	Predicted as Positive	Predicted as Negative
Positive	0.97001071	0.02998929
Negative	0.57043359	0.42956641

The updated confusion matrix shows a marked improvement in the True Positive Rate (TPR) at marginal cost of lower True Negative Rate (TNR). But at the same time, False Negative Rate (FNR) has improved drastically with a corresponding marginal increase in FPR. In this case, the dataset has more good customers than default ones (or bad customers), which makes it imbalanced. So, when working with imbalanced data, it is better to look at Recall (known as TPR), which minimizes FNR. Then, the credit scores and expected approval and rejection rates at the threshold is calculated based on following formulas in (5), (6):

$$approval\ rate = \frac{approved\ applications}{all\ applications} \quad (5)$$

and

$$rejection\ rate = 1 - approval\ rate \quad (6)$$

Based on the dataset, this threshold has a good approval rate of 0.926343, rejection rate of 0.073657, and the score of 488. However, the score at this threshold is relatively low in comparison to the accepted score range of some financial institutions in Vietnam like FE Credit (only accept leads with credit score above 630). Therefore, further, businesses can manually tweak the score cut-off based on the customer demographics of each bank and FIs for the highest business efficiency.

V. CONCLUSION

This work has proposed and designed a credit scoring and Probability of Default (PD) model using machine learning algorithm. Based on the experimental results, Logistic Regression returns higher accuracy. Thus, the model is determined to be built upon the Logistic Regression algorithm, and each feature in the consumer's profile is a factor in the score calculation. The chosen scorecard model is a common model used for credit appraisal in banking, and Weight of Evidence (WoE) is the method to find the necessary features based on the ranking of Information Value (IV). Moreover, setting a threshold to fit the customer demographics of each bank or FIs is also proposed to help

increase loans approval rate and enhance credit risk control flexibility.

On the basis of the obtained results, the research can be developed in the following directions:

- Evaluate profiles and scores of bad customers to propose features that need improvement to give them the opportunity to approach loans with low interest.
- Analyze and add features related to different types of income, especially in the form of cash transfers to the dataset.

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